

TCS, AI, and the Adults in the Room

Clément Canonne

October 2026

When I was a kid, my parents looked in control. Even when bad things happened, they knew what to do, what to say, told us not to worry, did the things adults do to make things work out, and things did work out.

Now that I am less of a kid, I realize that they were not (always) in control. When bad things happened, they took responsibility to *reclaim* agency. They put on an adult face, figured out what to say, told us not to worry, *and then silently did whatever they could so that things worked out*. They just did their best, and we did not know, and we did not have to worry.

And now, for something completely different. AI. We're currently facing a rather obvious crisis, "what happens when a self-proclaimed unstoppable force meets a historically very slow-moving object." I'm not going to detail what you probably all have witnessed and seen evolve over what feels like an eternity and the blink of an eye, but *we need to reevaluate many things*. Bad things are happening.¹ We are *not* in control. We do not really know what to do. But, as Omer Reingold wrote so well,² if you are a senior academic or industry researcher, **snap out of it**:

"Contemplating the end of your profession from the relative comfort of an established, tenured career is a privilege, and it comes with responsibilities. Senior academics are not merely individual researchers; we are stewards of our field, and we owe our junior colleagues active leadership rather than abandonment."

Perhaps more bluntly, and less eloquently than Omer put it: **we "senior academics" are supposed to be the adults**. Complacency, doom-erism, and generally navel-gazing, while tempting, are unacceptable, counter-productive, and frankly unbecoming: we have a duty of care to our students, mentees, our community.

We need to put on an adult face, figure out what to say to them, tell them not to worry, *and then silently do whatever we can so that things work out for them*.

That's not to say we should lie! It is, again, about taking responsibility, and coming up with a plan and solutions: letting others know that we are here for them, and that they can count on us. So I'll outline some thoughts here about what to do now: mostly two. (1) We must find out what we want to fight for and preserve. (2) One concrete action to take.



¹Some good ones, too.

²If you haven't, go read his post on the Theory Dish: <https://theorydish.blog/2026/09/29/speeding-up-the-process-of-mourning/>.

What's below is mostly targeted at "senior people," but feel free to read on if you are junior – Ph.D., postdoc, untenured faculty or (good god!) born after *The Dark Knight* was released in theaters.

(1) We need to figure out what our values are, our justification, and we need to do it yesterday.

So let's all take a break from freaking out about the latest longstanding conjecture whose resolution just appeared on the arXiv, and think about what it means to be doing Theoretical Computer Science, what our social mandate is, *why* we are doing what we are doing and *why* society has been paying us (quite well, actually) to do so.

- Is it to create knowledge, investigate fundamental questions for the very sake of it?
- Is it to disseminate knowledge, practical and/or abstract, to society?
- Is it to equip students, undergraduate and postgraduate, with a specific set of skills, to view and navigate the world with a computational mindset?
- Is it to solve problems relevant to society and to industry, with the bulk of our research being in some sense a "side quest" in support of this goal?
- ...
- Is it to keep us at arms' length, and prevent a bunch of people with an unhealthy knack for solving math puzzles from doing much worse things, like building an underground lair or starting a financial crisis?

Or is it something else entirely? Or a combination of all of the above? We may all have slightly different views on this, and I don't have definite answers, but *we should all be thinking about it* (or, rather, we should all *already* have been thinking about it). Thankfully, we don't have to start from nothing, because the giants whose shoulders we are camping on *did* give it some thought, and still are (see, e.g., [Aho+96], and the response [GW96] for a start³): but we must, starting from there, form our own opinions, decide what our values are, why we do what we do, and why society should give us a license to do so. **We won't know what to do, can't figure out what to say, unless we know why we do what we do and what we stand for as a community.** Putting my money where my mouth is, here are some of my own views.⁴

- (a) *We are meant to create knowledge, build theories, and new ways to look at things* This involves many different aspects, among which "proving theorems" is at best the tip of the iceberg (unfortunately, as for icebergs, this is also usually the only one people notice). We should explore what "computation" means *and could mean*, driven by curiosity, yes, but also because it reveals new understanding which can shape the way we, and down the line many others, think. I see the concept of *reduction between problems* and the shift in thinking associated with it as one of the most beautiful and influential contributions of TCS: one which not only influenced (deeply) the theory of computation, algorithms, and complexity, but also one we teach our students, and might change the lens through which they view other aspects of their life.
- (b) *We are not mathematically-inclined tech consultants.* As theoretical computer scientists, our *raison d'être* is not to solve problems from or for industry, or to only worry about current or near-future practical tasks they want to solve.⁵ This is not to say we shouldn't think about these problems as a source of potentially interesting research questions. Interesting algorithmic questions can come out of anywhere, and industry has often happened to be a good source! But "you are not solving a practical problem" is not a valid criticism of TCS, much like "you are not building fridges" is not a damning indictment of thermodynamics.

³And a blog post by Lance Fortnow providing some context and perspective here: <https://blog.computationalcomplexity.org/2016/06/karp-v-wigderson-20-years-later.html>.

⁴As I said, I don't have definite answers: if you threw a rock in the general direction of the TCS community, you'd likely hit someone (i) with slightly different opinions, (ii) looking furious about a recent bump on their head.

⁵There is a counterpart, though: we should not "claim practical relevance" if there is none. We are not beholden to solving industry questions, but *if* we claim to do so it has to be true. Cake, eating, having it too!

- (c) *We are meant to create knowledge, but that means disseminating it.* The license society gives us for our curiosity-driven endeavours and theory building comes with a price:⁶ we need to then explain this knowledge, communicate and teach it to our peers, our research community, our graduate students, and have it percolate through our teaching for it to reach as many people as possible. Knowledge creation without efforts towards exposition and dissemination is just a self-centered endeavour, with no benefit to the research community or society, and no place in academia.

And if you're wondering where to have these discussions, and how to be able to debate and exchange ideas in a fragmented world, I have good news (or at least a suggestion): StackExchange! CSTheory.SE (<https://cstheory.stackexchange.com/>) exists; it can be used; *let's use it!*

That was the first point. The second one, equally important: *what can we do to make things work for our students, postdocs, and the new generation?* Again, I don't have answers, just one big suggestion: **let's shield them during this chaotic transition.**

(2) Look at whatever grant you have, or can apply for, put funding together, call every possible philanthropist or funding body you can, be creative: let's offer as many 2-, 3-, 4-year fellowships, postdoc, or early-career researchers positions as possible for juniors to weather the storm.

Let me be clear: if you are tenured (or have a similarly stable position), now is not the time for hand-wringing. But a 3-year postdoc position which includes part-time teaching (assuming, as I do, that great teaching and exposition will always be in demand) or some side industry experience? This could make the difference between a promising researcher leaving the field, or managing to make it through and adapt to the changes our discipline is going through. A small endowment from some startup or industry? This could tip the scales and provide stability just for the few months a PhD student needs before landing a job they want. So now is the time to try our very best, and, without publicly lamenting the end of our field (or, worse, rejoicing in it), to **silently do whatever we can to so that things work out for the new generation.**

And of course, do take me up on it. I'll be trying, too (if I don't, you know where to find me. There's *probably* a very short flight from where you are to Sydney, and my office is always open.)



References

- [Aho+96] Alfred V Aho et al. "Theory of computing: Goals and directions". In: *Special Report of the National Science Foundation of the USA* (1996) (cit. on p. 2).
- [GW96] Oded Goldreich and Avi Wigderson. "Theory of Computing: A Scientific Perspective". In: *ACM Comput. Surv.* 28.4es (1996), p. 218 (cit. on p. 2).

⁶It's a small price, but one we tend not to focus on unfortunately. For several reasons, among which that it's typically less explicitly asked of us, and while it does take a lot of time it comes with few incentives.